

Bernoulli's Principle — The Flying Ball





The Big WOW! Flying Ball Magic

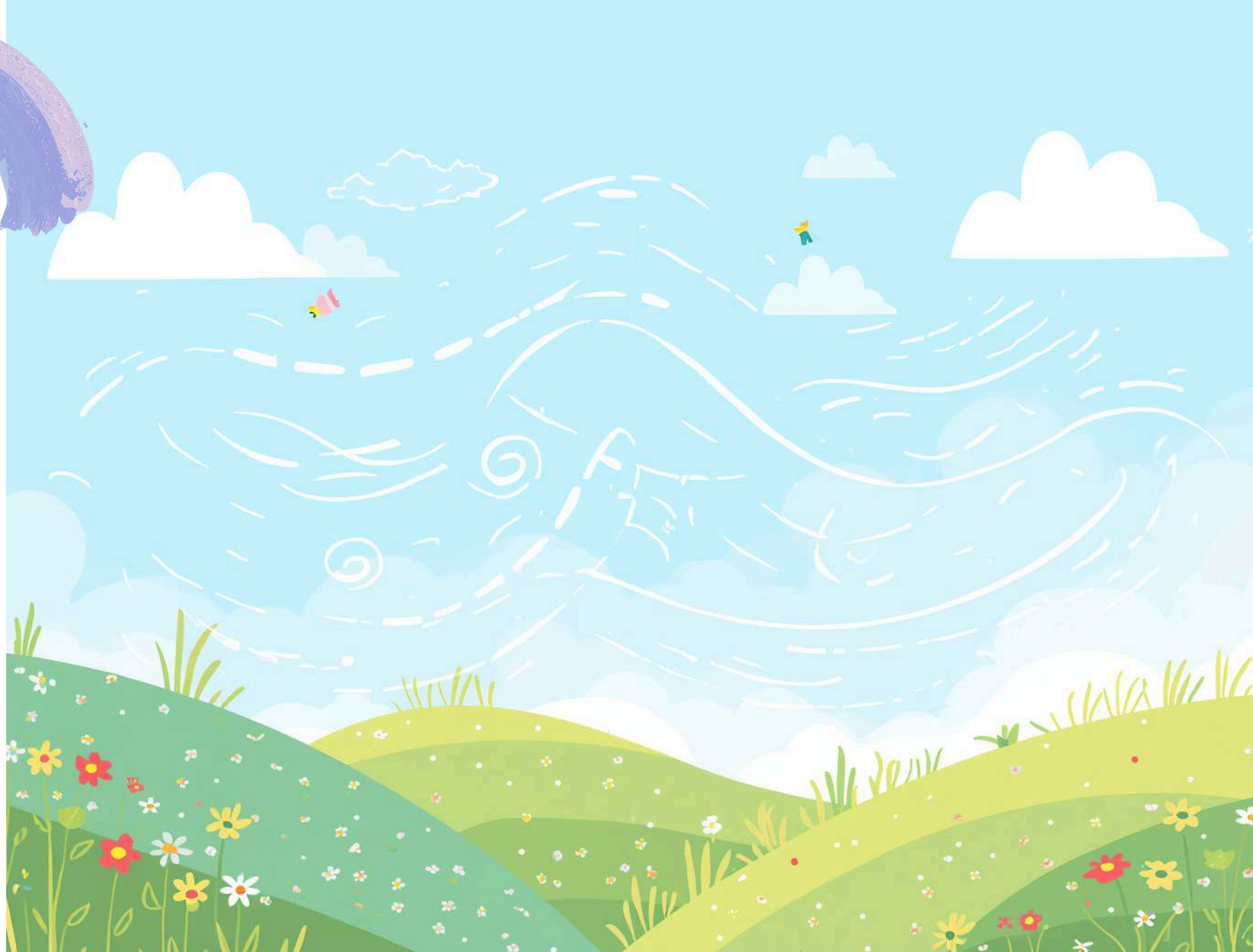
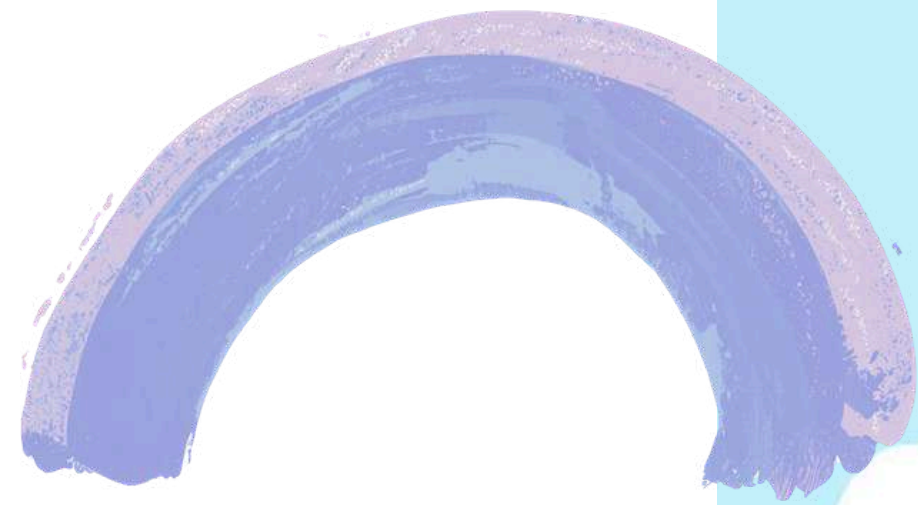
Yeh magic hai? Nahi! Yeh SCIENCE hai!

Watch the ball float in the air like magic — but it's all science!

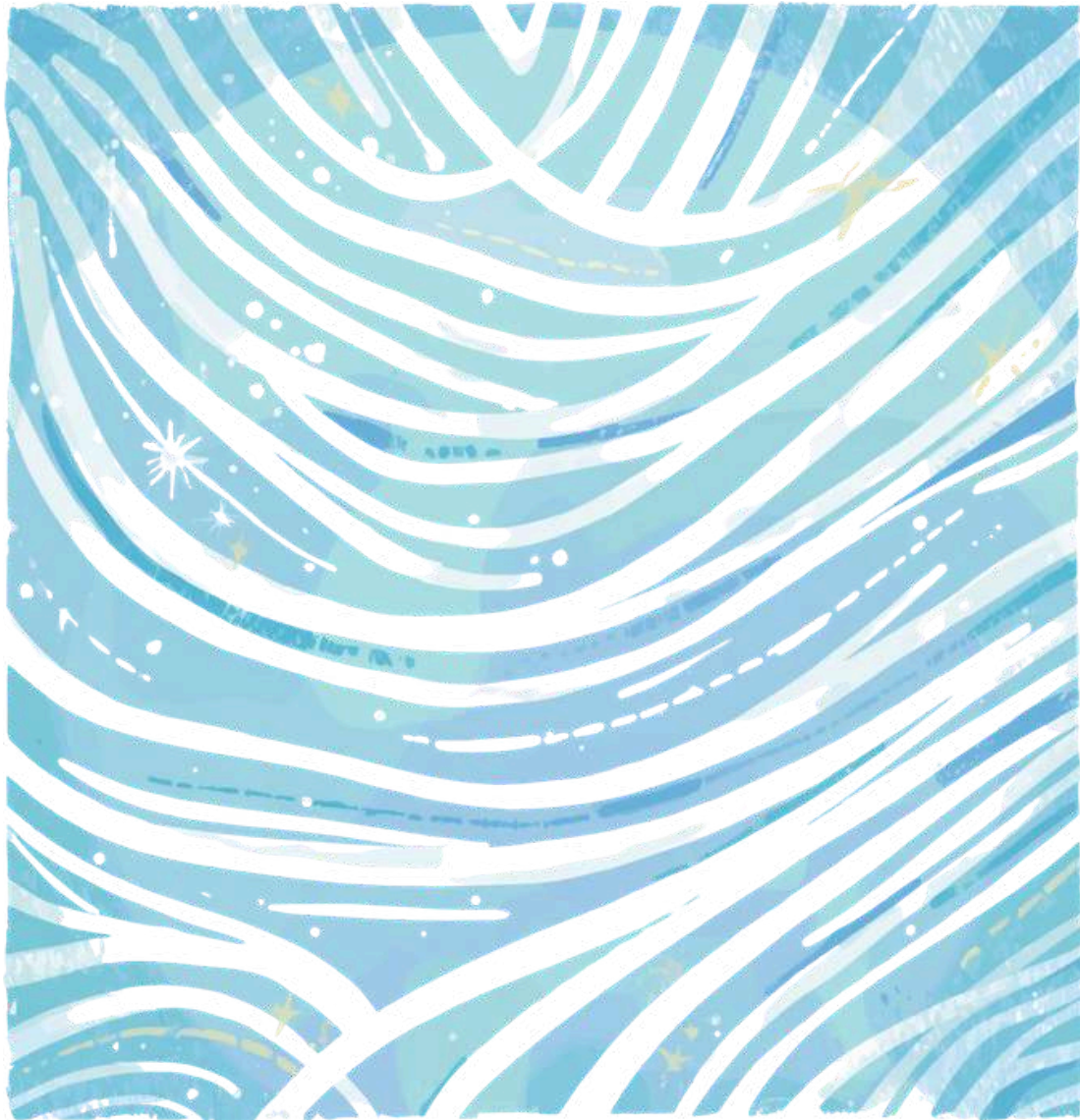
Did you know blowing air can make a ball FLOAT?

Fast moving air has a special power — let's discover Bernoulli's secret!

**Air is
Real!**



Air is All Around Us



Kite Flying Fun!

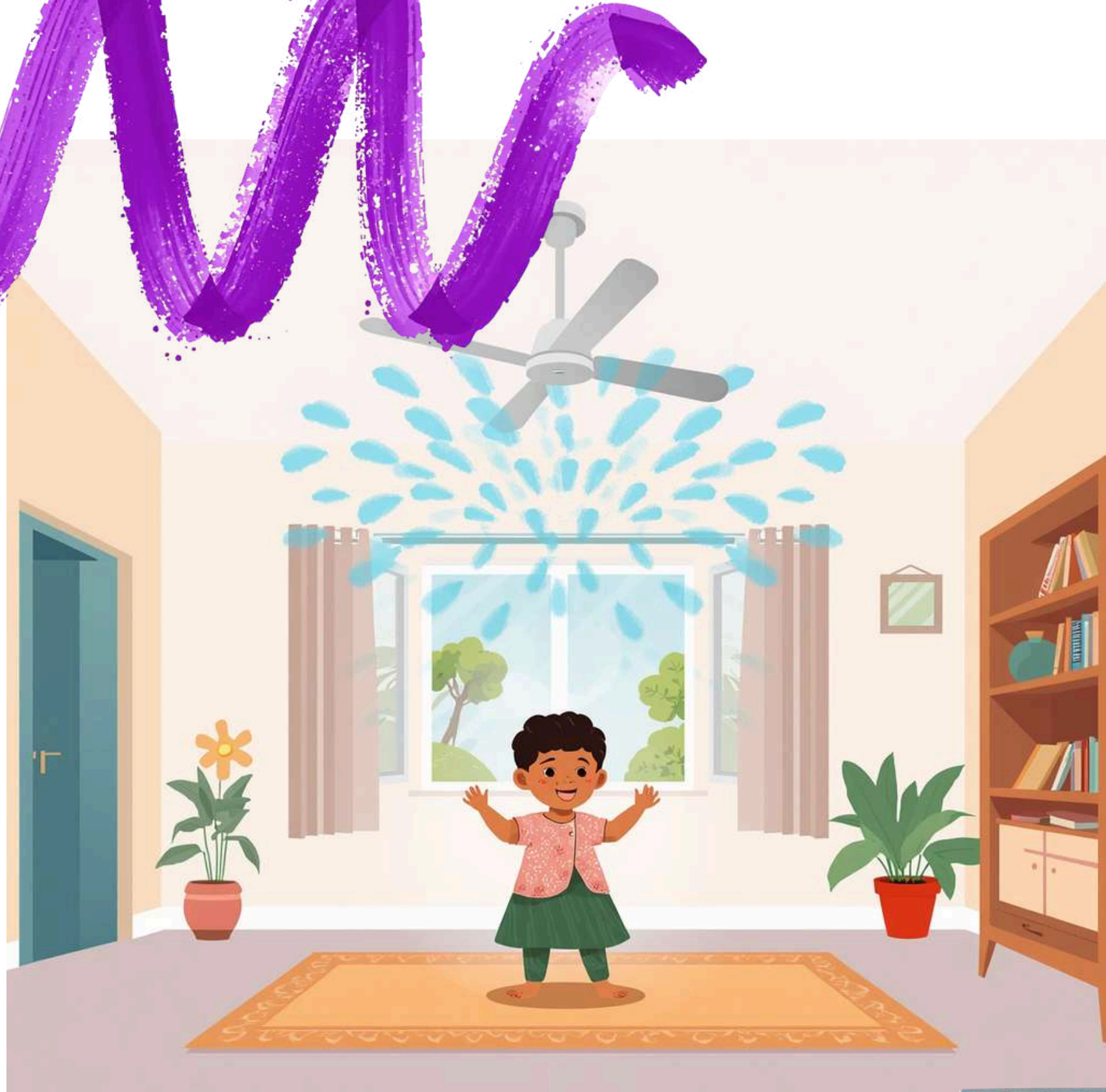
On Makar Sankranti, wind pushes our colorful kites high into the sky!

Clothes Drying Magic

On windy days, air pushes and dries our clothes on the terrace!

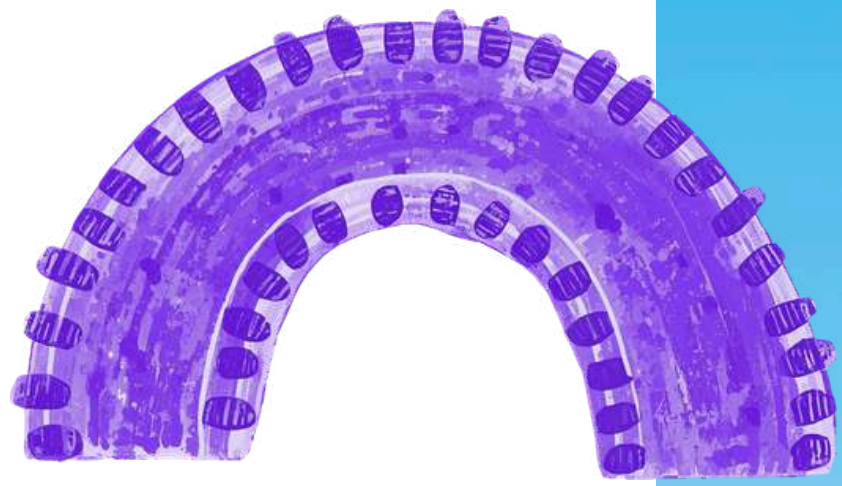
Cooling Hot Chai

When we blow on hot chai, air pushes the heat away to cool it down!

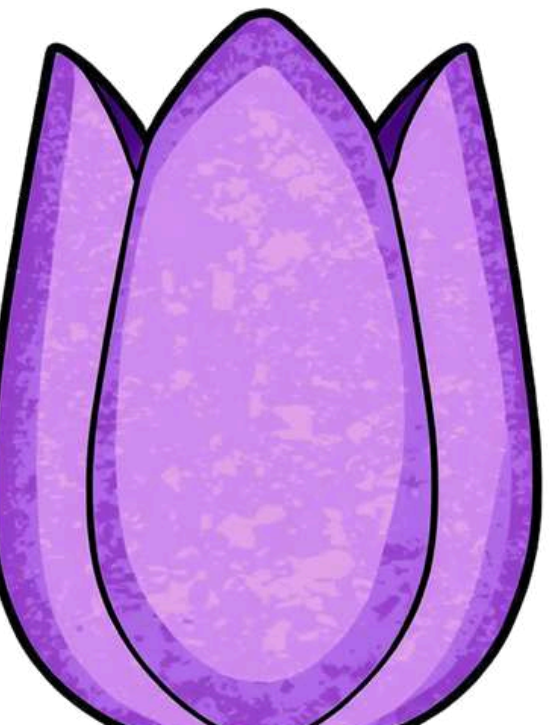


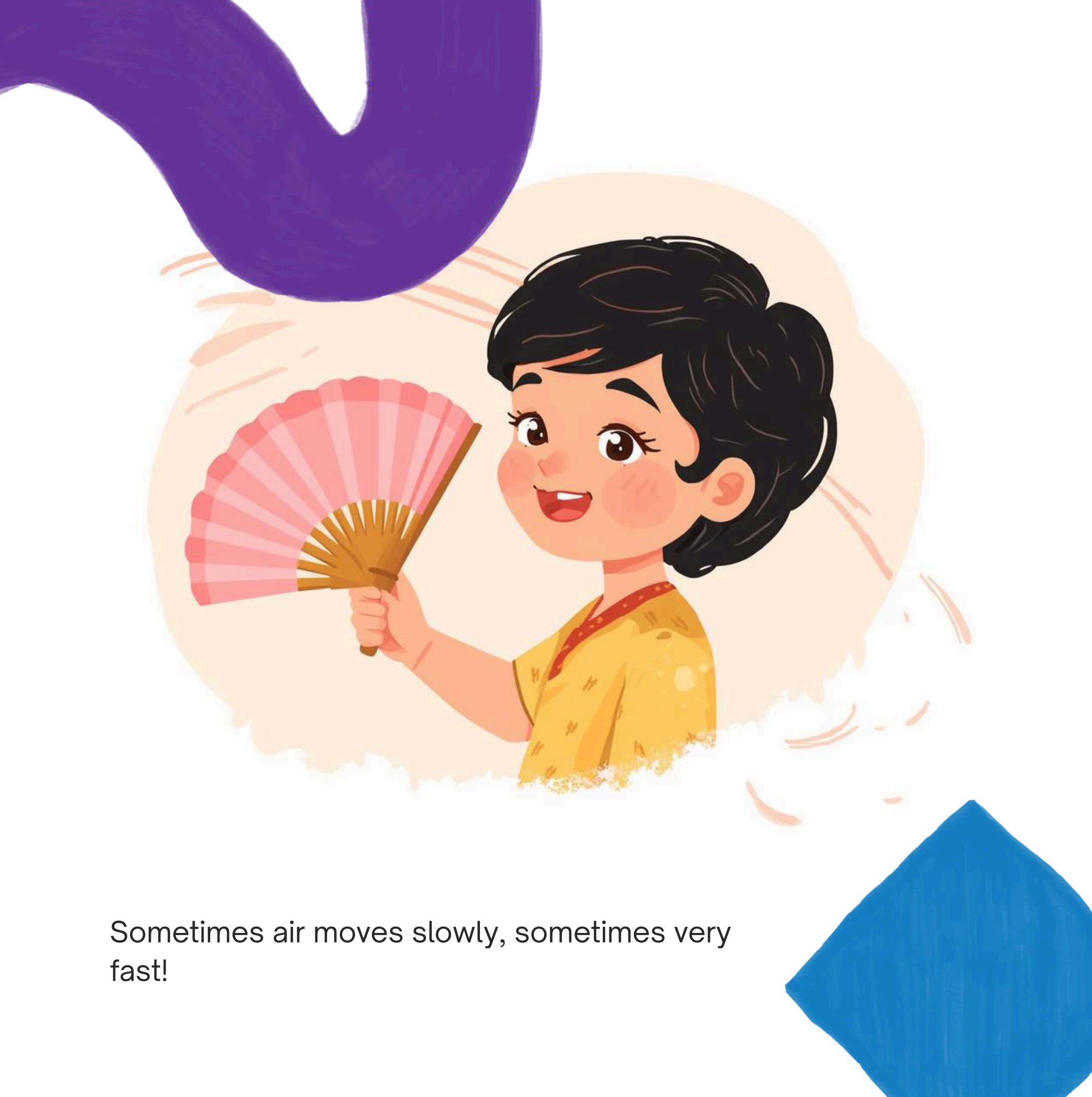
Air pushes when it moves!





Air Moves at Different Speeds





Sometimes air moves slowly, sometimes very fast!

Indian Examples of Air Speeds



Calm Summer Breeze

A gentle evening breeze cools us down slowly and softly.

Monsoon Wind Gusts

Strong monsoon winds blow fast and push things around!

Blowing on a Feather

Blow gently — feather floats.
Blow hard — it flies away fast!



Bernoulli's Big Idea: Fast Air Pushes Less



The Busy Street Analogy



Fast Air = Less Pushing

Fast air is like cars zooming on a highway — they move quickly and push less!

Slow Air = More Pushing

Slow air is like a traffic jam — cars are stuck and push more! Say it: BER-NOO-LEE'S PRINCIPLE!





Cricket Example — Swing Bowling

Bernoulli in Action!

Fast air moves differently on ball sides and makes it swing! That's Bernoulli in cricket!

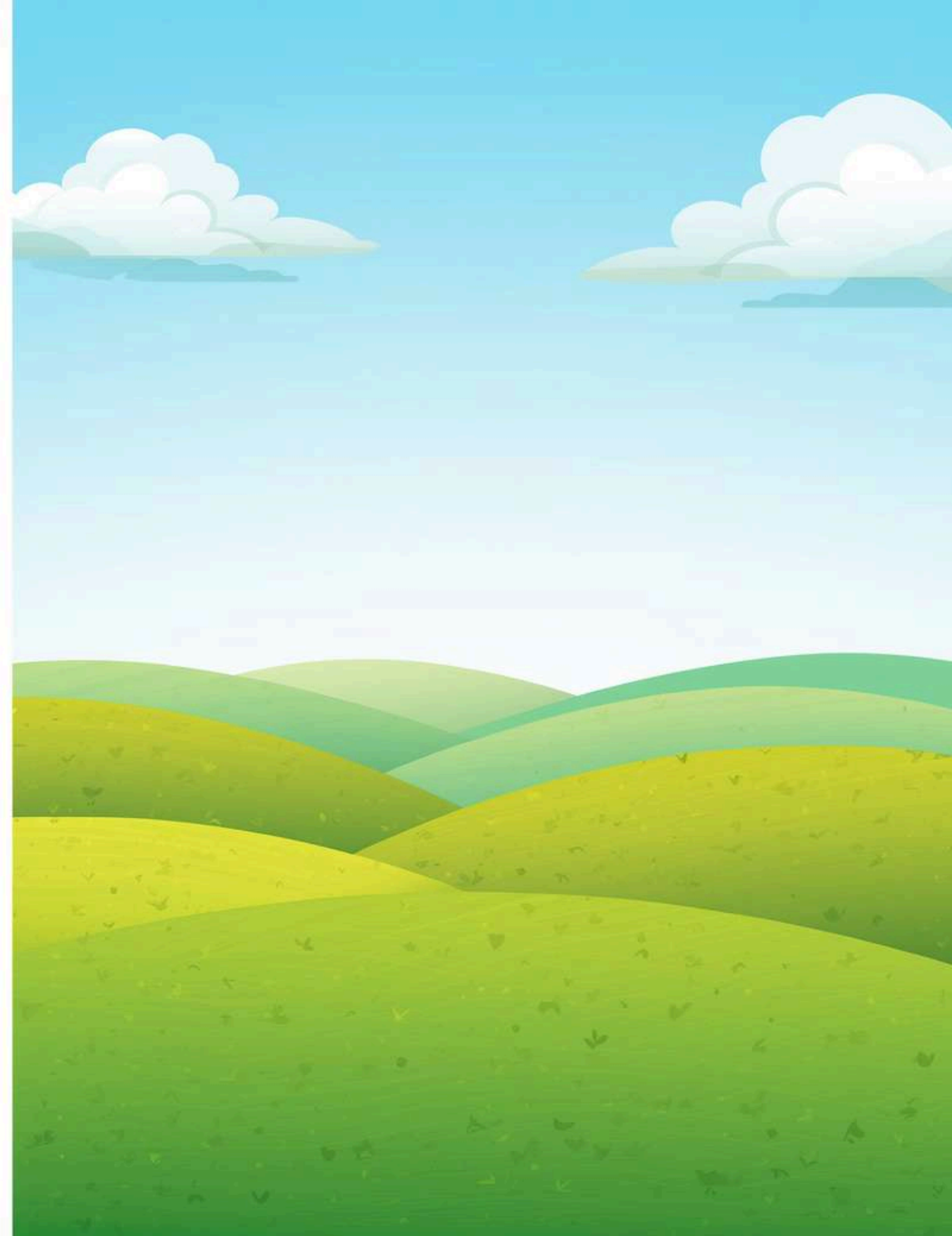
Fun Fact!

Cricket is India's favourite sport!





The Flying Ball Experiment





How the Ball Floats

Fast Air Pushes LESS!

The hair dryer blows fast air. Fast air pushes less than slow air around it!

Slow Air Pushes Back!

Slow air outside pushes the ball back inside. That's why the ball stays floating!

Try This At Home!



Step 1: Make a Ball

Crumple newspaper into a small ball. Get a straw and a small bowl (katori).

Step 2: Set It Up

Place the newspaper ball inside the bowl. Hold your straw ready!

Step 3: Blow & Watch!

Blow air through the straw across the bowl top. Watch the ball float and move! Chalo! Let's play with air!



**Bernoulli is
Everywhere!**





Bernoulli is Everywhere!

Cricket Swing Bowling

Fast air on one side makes the ball curve and swing!

Kites on Makar Sankranti

Wind lifts colourful patangs high into the sky!

Airplane Wings

Fast air above wings helps planes fly up, up, up!

Ceiling Fan Blades

Spinning blades push cool air down to keep us comfy!



Recap + Quiz + Badge Award

What We Learned!

- Air is real and pushes things
- Fast air pushes LESS than slow air
- This is Bernoulli's Principle (Bernoo-lee)

Quiz Time!

Which air pushes more? Slow or fast? Why does the ball float?

You Did It!

BERNOULLI EXPLORER
Well done! You are a Bernoulli Explorer!

Thank You!
Jai Hind!

